

Problem 2

Write a recursive MySQL procedure compute the factorial of a number .

```
mysql> SET @@SESSION.max_sp_recursion_depth = 255;
Query OK, 0 rows affected (0.00 sec)

mysql> SHOW PROCEDURE STATUS WHERE Db = 'dbms_lab';
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Db      | Name           | Type      | Definer      | Modified      | Created      | Security_type | Comment      | character_set_client | collation_connection | Database Collation |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| dbms_lab | GetCustomerLevel | PROCEDURE | root@localhost | 2026-08-12 10:56:48 | 2026-08-12 10:56:48 | DEFINER      |              | cp850                | cp850_general_ci    | utf8mb4_0900_ai_ci |
| dbms_lab | GetStudents      | PROCEDURE | root@localhost | 2026-08-12 10:56:04 | 2026-08-12 10:56:04 | DEFINER      |              | cp850                | cp850_general_ci    | utf8mb4_0900_ai_ci |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> CALL Factorial(5, @result);
```

```
mysql> SET @@SESSION.max_sp_recursion_depth = 255;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE Factorial(
    ->     IN n INT,
    ->     OUT result BIGINT
    -> )
    -> BEGIN
    ->     IF n <= 1 THEN
    ->         SET result = 1;
    ->     ELSE
    ->         CALL Factorial(n - 1, @temp);
    ->         SET result = n * @temp;
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SET @@SESSION.max_sp_recursion_depth = 255;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE Factorial(
    ->     IN n INT,
    ->     OUT result BIGINT
    -> )
    -> BEGIN
    ->     IF n <= 1 THEN
    ->         SET result = 1;
    ->     ELSE
    ->         CALL Factorial(n - 1, @temp);
    ->         SET result = n * @temp;
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DELIMITER ;
mysql> CALL Factorial(5, @result);
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT @result;
+-----+
| @result |
+-----+
|      120 |
+-----+
1 row in set (0.00 sec)
```

